

JULIANN GERACI

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Department of Mathematics ◊ University of Nebraska - Lincoln

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EDUCATION

Doctor of Philosophy , Mathematics <i>University of Nebraska-Lincoln, Lincoln, NE</i>	Expected May 2026
Master of Science , Mathematics <i>University of Nebraska-Lincoln, Lincoln, NE</i>	December 2021
Bachelor of Arts , Mathematics <i>State Univeresity of New York at Oswego, Oswego, NY</i>	May 2020

POSITIONS HELD

Graduate Teaching Assistant <i>Mathematics Department, University of Nebraska- Lincoln</i>	August 2020 -
Graduate Research Assistant <i>Center for Science, Mathematics and Computer Education, University of Nebraska- Lincoln</i>	June 2021 - August 2021, June 2023-August 2023
Worked on the AIR@NE project, an NSF-funded grant that examines the adaptation and implementation of a validated K-8 Computer Science curriculum in diverse school districts.	
NSF Graduate Student Mentor <i>The Polymath Jr Program, City University of New York</i>	June 2023 - August 2023
NSF Research Assistant <i>REU, East Tennessee State University, Johnson City, TN</i>	June 2019- August 2019

PUBLICATIONS

1. *Adjoints of Morphisms of Codes* (with A.B. Kunin, A. Seceleanu), <https://arxiv.org/abs/2603.10837>.
2. *Boolean Matrix Rank via Monomial Ideals* (with A.B. Kunin, A. Seceleanu), <https://arxiv.org/abs/2509.09570>.
3. *Products and powers of principal symmetric ideals* (with E. Dannetun, B. Fang, R. Formenti, B. Gao, R. Kogel, Y. Li, S. Mandal, V. Rupasinghe, A. Seceleanu, D. Tran, N. Walker), *Journal of Algebra and Its Applications*, in press.
4. *Graphical Universal Cycles of Combinatorial Objects* (with A. Cantwell, A. Godbole, and C. Padilla), *Advances in Applied Mathematics*, Volume 127, June 2021, 102166.

AWARDS AND RECOGNITIONS

AWM Outstanding Graduate Student Research Poster <i>Joint Mathematics Meetings</i>	2026
Selected to present a research poster at the Joint Mathematics Meetings and awarded Best Poster in the Graduate Student Research Poster Session, recognizing excellence in originality, clarity of exposition, and impact of the research.	

MOST Fellow 2024

National Museum of Mathematics

Selected as one of a small national cohort for my ability to communicate complex mathematical ideas clearly and engagingly. Through this fellowship, I refined my public-facing presentation skills—focusing on precision, accessibility, and personal style—and delivered outreach talks to middle and high school girls to help broaden participation in mathematics.

Don Miller Outstanding Teaching by a Graduate Student 2022

University of Nebraska - Lincoln

Awarded annually to one graduate teaching assistant whose teaching demonstrates exceptional clarity, organization, and student impact. I received this award in recognition of my commitment to student success, my strong course evaluations, and my development of effective, active-learning teaching practices.

PROFESSIONAL DEVELOPMENT

**Advanced Studies Institute in Mathematics of
Data Science & Machine Learning**

Urgench State University (Uzbekistan)

January 2024

Neural Coding and Combinatorics

Institute for Computational and Experimental Research in Mathematics

October 2023

Commutative Algebra Market Preparation workshop

University of Nebraska-Lincoln

August 2023

Graduate Student Orientation Leader

University of Nebraska-Lincoln

2022-2023

Teaching and Learning Mathematics at the Post-Secondary Level I and II

University of Nebraska-Lincoln

2021-2022

A three-credit sequence of courses intended to train mathematics graduate teaching assistants how to be an effective teacher and classroom manager.

TEACHING EXPERIENCE

Associate Convener

- Math 100A: *Intermediate Algebra*

Fall 2022, Spring 2023

Instructor of Record

- Math 100A: *Intermediate Algebra*
- Math 300: *Mathematics Matters*
- Math 101C: *College Algebra Co-requisite*
- Math 314: *Linear Algebra*
- Math 301: *Geometry Matters*

Fall 2021, Spring 2022, Fall 2022, Spring 2023

Spring 2024

Fall 2024

Summer 2025

Spring 2026

Recitation Instructor

- Math 106: *Calculus I Recitation*

Fall 2020, Spring 2021 (2 sections),
Summer 2021 (1 section), Fall 2023 (3 sections),
Fall 2025 (2 sections)

Teaching Assistant

- Math 435: *Math in the City*
- Math 812T: *Geometry for Geometry Teachers*

Fall 2025, Spring 2026

Summer 2022

Curriculum Development

- Course Repository Summer 2024
Established a new repository dedicated to course information which will facilitate efficient communication and centralized access for instructors.
- Math 101C: *College Algebra Co-requisite* Summer 2022

Tutoring

- UNL Math Resource Center Fall 2020-
Drop-in tutoring center for students in general mathematics courses. Tutored two hours per week, one semester per year.

Grading Assignments

- Math 415/815: *Theory of Linear Transformations* Spring 2023

TALKS

19. **Neural Rings and Applications to Boolean Matrix Factorization** January 6, 2026
Washington D.C., JMM 2026
18. **Neural Rings and Applications to Boolean Matrix Factorization** October 4, 2025
Tulane University, AMS 2025 Fall Southeastern Sectional Meeting
17. **Boolean Rank via Monomial Ideals** October 4, 2025
Tulane University, AMS 2025 Fall Southeastern Sectional Meeting
16. **Boolean Rank via Monomial Ideals** May 17, 2025
University of Wisconsin-Madison, AWM Research Symposium
15. **Neural Rings and Ideals** April 11, 2025
SUNY Oswego, Alumni Series
14. **Boolean Rank via Monomial Ideals** April 6, 2025
University of Connecticut, AMS 2025 Spring Eastern Sectional Meeting
13. **Boolean matrix rank and Castlenuovo-Mumford regularity**(Poster) March, 2025
Fields Institute
12. **How Algebra Can Help Prevent Theft** September 12, 2024
University of Nebraska-Lincoln, Commutative Algebra Seminar
11. **Neural Rings and Ideals** November, 2024
Grinnell College, Great Plains Alliance Series
10. **Boolean matrix rank and Castlenuovo-Mumford regularity**(Poster) August 15, 2024
University of Notre Dame, UweFest
9. **Simplicial Resolutions and the Scarf Complex** September 20, 2023
University of Nebraska-Lincoln, Commutative Algebra Seminar
8. **Monomial Resolutions** April 26, 2023
University of Nebraska-Lincoln, Commutative Algebra Reading Seminar
7. **Introduction to Neural Codes, Rings, and Ideals** March 14, 2023
Dordt College, Great Plains Alliance Series
6. **Neural Rings and Ideals** December 5, 2022
Online, Commutative Algebra Regional Expository Seminar
5. **Neural Rings and Ideals** November 30, 2022
University of Nebraska-Lincoln, Commutative Algebra Reading Seminar

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| 4. A Path to Resolutions
<i>University of Nebraska-Lincoln, Commutative Algebra Reading Seminar</i> | August 31, 2022 |
| 3. Gröbner Bases II
<i>University of Nebraska-Lincoln, Commutative Algebra Reading Seminar</i> | March 9, 2022 |
| 2. Gröbner Bases I
<i>University of Nebraska-Lincoln, Commutative Algebra Reading Seminar</i> | March 2, 2022 |
| 1. Construction of Free Resolutions Through Simplicial Complexes
<i>University of Nebraska-Lincoln, Commutative Algebra Reading Seminar</i> | October 6, 2021 |

SERVICE

NCUWM Committee Member 2024 -
University of Nebraska - Lincoln

The Nebraska Conference for Undergraduate Women in Mathematics is an event dedicated to promoting women's research in math by giving them the opportunity to network and present at a conference.

Girls Inc. 2024
University of Nebraska - Lincoln

The Eureka program by Girls Inc. of Lincoln gives 8th–12th grade girls the chance to explore STEM through hands-on activities, including doing math in a supportive, pro-girl environment.

UNL Math Circle 2023-
University of Nebraska - Lincoln

Assists with outreach sessions where university mathematicians engage high school students in activity-based explorations of advanced mathematical topics. Emphasizes problem solving beyond the standard high school curriculum.

Great Plains Alliance 2022-
University of Nebraska - Lincoln

Gives outreach talks to undergraduates at regional colleges through the Great Plains Alliance. Participates in Q&A sessions about graduate school and careers in mathematics.

Commutative Algebra Reading Seminar Co-Organizer 2022-2023
University of Nebraska - Lincoln

Co-organized a weekly seminar for graduate students to present and discuss topics in commutative algebra. Facilitated scheduling, paper selection, and group discussions.

Graduate Student Seminar Co-Organizer 2021-2022
University of Nebraska - Lincoln

Co-led a weekly graduate student seminar featuring talks on research and mathematical topics. Promoted communication, peer learning, and presentation practice.

NCUWM Volunteer 2020 -
University of Nebraska - Lincoln

Supports the Nebraska Conference for Undergraduate Women in Mathematics by assisting with logistics, talk timing, and participant support.

UNL Math Day Volunteer

2020 -

University of Nebraska - Lincoln

Volunteers at an annual event for over 1,500 high school students from across Nebraska. Helps run competitions and guides students through math-related activities.

COMPUTER SKILLS

Languages/Software: C++, Git/GitHub, Google Workspace, HTML/CSS, Java, JavaScript, Julia, LaTeX, Macaulay2, MatLab, Mathematica, Microsoft Office, Python

Algorithms: Discrete Fourier Transform, Euler Method, Gauss-Seidel Method, Gaussian Elimination, Jacobi Method, Newton Method, Runge-Kutta Methods, Steepest Descent